FE Graded Quiz M7

Question 1

Which of the following is **not** a classification model?

- Decision tree
- Na√Øve Bayes classifiers
- Principal component analysis
- K-nearest neighbors

Question 2

For a VAR(3) model with 5 variables calculate the number of parameters that need to be estimated (including constants).

- 15
- 90
- 80
- 75

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Question 3

Design a research approach to investigate potential spurious Granger causality relationships in a network of 10 financial assets. Which of the following would be most effective?

- Reduce the lag order to minimize overfitting
- Increase the significance level to reduce Type II errors
- Run only bivariate Granger causality tests between all pairs
- Use a multivariate VAR framework and control for common factors

Question 4

According to the following factor analysis result, which factors should be included for analysis?

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	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
SS Loadings	7.3	5.2	2.1	0.8	0.7	0.65
Propotion Var	0.304	0.258	0.1	0.05	0.02	0.01
Cumulation Var	0.304	0.562	0.662	0.712	0.732	0.742

- All of them
- Factor 1, Factor 2
- Factor 1, Factor 2, Factor 3
- Factor 1

Question 5

Two financial assets have a partial correlation of exactly zero after applying Graphical LASSO. What does this indicate about their relationship in the network context?

- The assets have a time-lagged relationship that wasn't captured in the model
- The assets have no correlation whatsoever and will never move together
- The assets are connected only through a common factor but not directly

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- The assets are conditionally independent given all other assets in the network

Question 6

If the original p-value threshold is 0.05 and we use the Bonferroni method to adjust for 30 simultaneous tests, what is the adjusted p-value threshold?

- 0.00833
- 1.5
- 0.00167
- 0.0015

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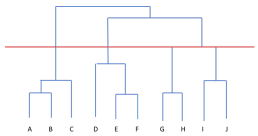
QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Question 7

Given the following dendrogram and the red cutoff line, how many clusters would be created?



- 4
- 3
- 6

TOP

- 5

Question 8

You observe that adding stock X's past values improves the prediction of stock Y's returns reducing RMSE from 0.045 to 0.042. The F-statistic is 3.2 with a p-value of 0.06. Given a 5% significance level evaluate the evidence for X Granger causing Y.

- Strong evidence that X Granger causes Y
- Inconclusive result requiring additional testing
- No evidence of Granger causality despite prediction improvement
- Weak evidence that X Granger causes Y but not statistically significant

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Question 9

If you found that X Granger causes Y what can you definitively conclude?

- X and Y have a common cause
- X directly causes changes in Y
- Y cannot Granger cause X
- Past values of X help predict future values of Y

Question 10

Which of the following statements regarding rotations in factor analysis is **false**?

- There are two types of rotation methods: orthogonal rotation and oblique rotation.

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- Varimax makes some loadings as large as possible while some loadings are very small in absolute value in order to make interpretation easier.
- Oblique rotation allows factors to be uncorrelated.
- Varimax is an orthogonal rotation.

Question 11

A researcher finds that a simple linear regression-based asset network shows all assets are bidirectionally connected, while adding S&P 500 as a confounder reveals that some pairs become conditionally independent. What financial interpretation best explains this phenomenon?

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- Linear regression is fundamentally unsuited for financial network analysis
- The S&P 500 index distorts the true relationships between individual assets
- The assets have no true economic relationships with each other
- The apparent direct connections between some assets were actually mediated by common exposure to market movements

Question 12

For an undirected asset network with 8 nodes, what would be the possible Bonferroni-adjusted p-value threshold if the original threshold is 0.01?

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- 0.01
- 0.000357
- 0.56
- 0.05

Question 13

Which of the following information is **not** necessary to estimate posterior conditional probability for LDA?

- Sample means for each type
- Correlation among different types
- Sample variance-covariance matrix for each type
- Prior probability for each type

Question 14

Calculate the optimal lag order for a VAR model if the following information criteria values

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were obtained: Lag 1: AIC = 2340 BIC = 2380 HQ = 2355 Lag 2: AIC = 2320 BIC = 2375 HQ = 2345 Lag 3: AIC = 2310 BIC = 2370 HQ = 2340

- Lag 3
- Lag 1
- Cannot be determined from the given information
- Lag 2

Question 15

When performing a Granger causality test what does the F-statistic compare?

- The mean of the squared residuals from the full and reduced models
- The sum of squared residuals from the full and reduced models

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- The ratio of the difference in RSS between models to the RSS of the full model adjusted for degrees of freedom
- The covariance between the two time series

Question 16

When we apply rotation to factor analysis, which of the following statements is **correct**?

- Loadings will stay the same, but the percentage of total variance of data explained will change.
- Loading will change, but the percentage of total variance of data explained will stay the same.

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- Neither loadings nor the percentage of total variance of data explained will change.
- Both loadings and percentage of total variance of data explained will change.

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